

SECR2043 OPERATING SYSTEMS

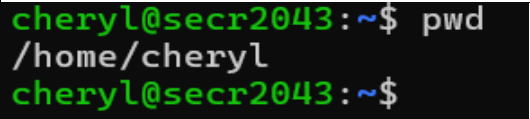
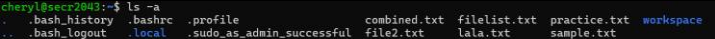
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 Student ID : A 2 3 C S 0 0 6 0 _____
 Section : 0 2 _____

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write **"No output"** instead.

1.	Display your current working directory.	
	Command	Output
	pwd	
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -a	
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	mkdir os_lab	No output
4.	Change your current working directory to "os_lab".	
	Command	Output
	cd os_lab	No output
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output
	touch hello.txt echo "Hello,world!" > hello.txt	No output

6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	cheryl@secr2043:~/os_lab\$ cat hello.txt Hello,world!
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	cp hello.txt hello_copy.txt	No output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	mv hello_copy.txt hello_bak.txt	No output
9.	Delete "hello.txt".	
	Command	Output
	rm hello.txt	No output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	ls -Ralh	cheryl@secr2043:~/os_lab\$ ls -Ralh .: total 12K drwxr-xr-x 2 cheryl cheryl 4.0K Apr 19 06:50 . drwxr-x--- 5 cheryl cheryl 4.0K Apr 19 06:38 .. -rw-r--r-- 1 cheryl cheryl 13 Apr 19 06:47 hello_bak.txt
Total Mark:		

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt	No Output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello World" >hello.txt	No Output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >>hello.txt	No Output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	<pre>cheryl@secre2043:~/os_lab\$ cat hello.txt Hello World This is a test</pre>
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	touch bye.txt echo "Goodbye World" >bye.txt	No Output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	<pre>cheryl@secre2043:~/os_lab\$ cat hello.txt bye.txt Hello World This is a test Goodbye World</pre>
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	cat hello.txt bye.txt >greeting.txt	No Output

8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	cheryl@secr2043:~/os_lab\$ cat greeting.txt Hello World This is a test Goodbye World
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt GNU nano 7.2 greeting.txt * Hello Cheryl This is a test Goodbye Cheryl	No Output
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	cheryl@secr2043:~/os_lab\$ cat greeting.txt Hello Cheryl This is a test Goodbye Cheryl
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	<pre>nano hello.c GNU nano 7.2 hello.c #include <stdio.h> int main() { printf("Hello World\n"); return 0; }</pre>	No Output
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	<pre>gcc hello.c -o hello</pre>	No Output
3.	Run the executable file "hello" and display its output.	
	Command	Output
	<pre>./hello</pre>	<pre>cheryl@secre2043:~\$./hello Hello World</pre>
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	<pre>nano sum.c GNU nano 7.2 sum.c #include <stdio.h> #include <stdlib.h> int main(int argc, char *argv[]) { if (argc != 3) { printf("Usage: %s <num1> <num2>\n", argv[0]); return 1; } int num1 = atoi(argv[1]); int num2 = atoi(argv[2]); int sum = num1 + num2; printf("Sum: %d\n", sum); return 0; }</pre>	No Output
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output

	gcc sum.c -o sum	No Output
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum 10 20	cheryl@secr2043:~\$./sum 10 20 Sum: 30
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum -5 15	cheryl@secr2043:~\$./sum -5 15 Sum: 10
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c <pre> GNU nano 7.2 echo.c #include <stdio.h> int main() { char input[100]; printf("Enter a line of text: "); fgets(input, sizeof(input), stdin); printf("You entered: %s", input); return 0; } </pre>	No Output
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc echo.c -o echo	No Output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	./echo Input = Hello World	cheryl@secr2043:~\$./echo Enter a line of text: Hello World You entered: Hello World
	Total Mark:	